Interference is a phenomenon in which two waves superimpose to form a resultant wave of greater, lower or the same amplitude.

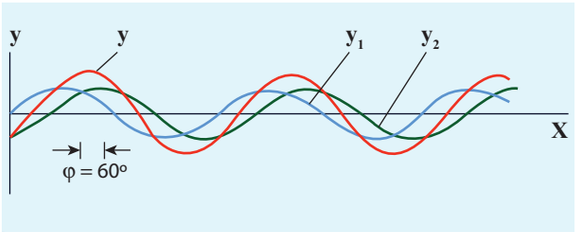


Figure 11.28 Interference of two sinusoidal waves

Consider two harmonic waves having identical frequencies, constant phase difference φ and same wave form (can be treated as coherent source), but having amplitudes A_1 and A_2, then

$$y_1 = A_1 \sin(kx - \omega t) \quad (11.47)$$

$$y_2 = A_2 \sin(kx - \omega t + \varphi) \quad (11.48)$$

Suppose they move simultaneously in a particular direction, then interference occurs (i.e., overlap of these two waves). Mathematically

$$y = y_1 + y_2 \quad (11.49)$$

Therefore, substituting equation (11.47) and equation (11.48) in equation (11.49), we get

$$y = A_1 \sin(kx - \omega t) + A_2 \sin(kx - \omega t + \varphi)$$

Using trigonometric identity $\sin(\alpha+\beta) = (\sin\alpha\cos\beta + \cos\alpha\sin\beta)$, we get

$$y = A_1 \sin(kx - \omega t) + A_2 [\sin(kx - \omega t)\cos\varphi + \cos(kx - \omega t)\sin\varphi]$$

$$y = \sin(kx - \omega t)(A_1 + A_2\cos\varphi) + A_2 \sin\varphi \cos(kx - \omega t) \quad (11.50)$$

Let us re-define

$$A\cos\theta = (A_1 + A_2\cos\varphi) \quad (11.51)$$

$$\text{and } A\sin\theta = A_2\sin\varphi \quad (11.52)$$

then equation (11.50) can be rewritten as

$$y = A\sin(kx - \omega t)\cos\theta + A\cos(kx - \omega t)\sin\theta$$

$$y = A\,(\sin(kx - \omega t)\cos\theta + \sin\theta\cos(kx - \omega t))$$

$$y = A\sin(kx - \omega t + \theta) \quad (11.53)$$

By squaring and adding equation (11.51) and equation (11.52), we get

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\varphi \quad (11.54)$$

Since, intensity is square of the amplitude ($I = A^2$), we have

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\varphi \quad (11.55)$$

This means the resultant intensity at any point depends on the phase difference at that point.

(a) For constructive interference:

When crests of one wave overlap with crests of another wave, their amplitudes will add up and we get constructive interference. The resultant wave has a larger amplitude than the individual waves as shown in Figure 11.29 (a).

The constructive interference at a point occurs if there is maximum intensity at that point, which means that

$\cos\varphi = +1 \Rightarrow \varphi = 0, 2\pi, 4\pi, \ldots = 2n\pi$, where $n = 0,1,2,\ldots$

This is the phase difference in which two waves overlap to give constructive interference.

Therefore, for this resultant wave,

$$I_{\text{maximum}} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2 = (A_1 + A_2)^2$$

Hence, the resultant amplitude

$$A = A_1 + A_2$$

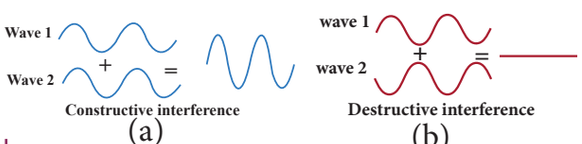


Figure 11.29 (a) Constructive interference (b) Destructive interference

UNIT-11(XI-Physics_Vol-2).indd 249 26-03-2022 17:24:52